AN2DL - First Challenge Report
NoNameTeam

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November 17, 2025

1 Introduction

This project addresses the domain of *multivariate time series classification* for predicting human pain levels. The dataset consists of temporal data collected from subjects.

The primary objective is to classify the actual pain status of the subjects into three distinct classes: [no_pain (0), low_pain (1), high_pain (2)]. Our goal is to develop a deep learning model that **accurately predicts these pain states**, assessed by the weighted **F1-score**. Our approach focuses on using Recurrent Neural Networks (RNNs) to effectively capture the temporal dynamics and interdependencies within the sensor data.

2 Problem Analysis

Our models are evaluated according to the **weighted F1-score**, which is the primary metric for this challenge. It is defined as:

$$F1_{weighted} = \sum_{c \in C} w_c \cdot 2 \cdot \frac{\text{precision}_c \cdot \text{recall}_c}{\text{precision}_c + \text{recall}_c} \quad (1)$$

This metric's sensitivity to class distribution makes handling data imbalance a critical aspect of the project.

The given dataset consists of 661 unique samples, each comprising a time series of 160 steps. Each step contains 38 features, including 31 continuous joint measurements, 4 pain surveys, and 3 categorical body characteristics. We began with a data inspection phase to identify potential outliers and uninformative features. The objective was to create a more representative dataset by eliminating noise and redundancy, which is a crucial step for improving the model's learning capacity.

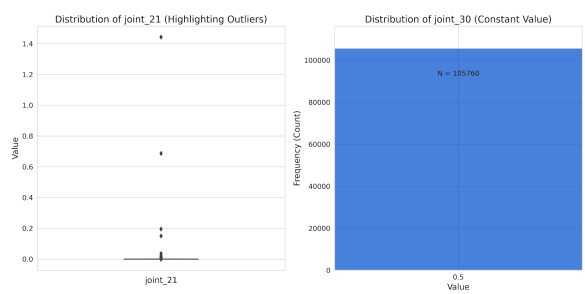

Figure 1: outliers and constant value columns.

This analysis revealed several key characteristics. We analyzed feature variance to identify uninformative data and found that the `joint_30` feature was constant across all samples, as visualized in Figure 1 (b), and was subsequently dropped from our feature set. We also noted that many categorical features showed almost no variance. Furthermore, as shown in Figure 1 (a), **some joint features exhibited significant outliers**, with maximum values far exceeding the main data distribution.

The most significant characteristic identified was a **strong class imbalance** in the full dataset, as shown in Table 1. The 'no_pain' class constitutes a vast majority (77.3%), while **both 'low_pain' and 'high_pain' are significantly underrepresented**. This imbalance directly guided our methodological choices to ensure the model learned to recognize the minority classes and not just predict the majority.

Table 1: Full training_set label distribution.

Class	Occurrences	Percentage
0: no_pain	511	77.31%
1: low_pain	94	14.22%
2: high_pain	56	8.47%
Total	661	100.00%

3 Method

Our methodology was structured to first address the dataset's specific characteristics through preprocessing, then systematically experiment with different models and techniques to overcome the problem's main challenges.

Data Preprocessing and Sequencing

We first transformed the raw tabular data into a format suitable for sequence modeling. This involved:

1. **Encoding**: Converting the categorical subject characteristics (`n_legs`, `n_hands`, `n_eyes`) into a suitable numerical representation.

2. **Scaling**: Applying clipping at the 99th percentile to the continuous sensor features (`joint_` and `pain_survey_`) to handle extreme outliers, followed by Min-Max scaling based on the training set statistics.

3. **Sequencing**: We used a sliding window to segment each subject's time series into overlapping sequences.

Class Imbalance

As noted, the dataset suffers from a significant class imbalance, with the *no_pain* class being far more prevalent than the others. To address this, we explored using a weighted loss function (equation 3), which modifies the standard Cross-Entropy Loss by applying a class-specific weight w_c to give higher importance to minority classes.

$$\mathcal{L} = -\frac{1}{N}\sum_{i=1}^{N} y_i \log(\hat{y}_i) \quad (2)$$

$$\mathcal{L}_{WCE} = -\frac{1}{N}\sum_{i=1}^{N} w_{c_i} \cdot \log(\hat{y}_i) \quad (3)$$

However, our final model performed best with the standard **Cross-Entropy Loss** (equation 2), relying more heavily on the model's architecture and strong regularization.

Validation and Hyperparameter Tuning

To ensure our model would generalize to unseen individuals rather than just memorizing patterns from subjects in the training set, we implemented a **subject wise validation strategy**. Instead of a simple random split, we used a **K-Fold Shuffle Split** to test different user based splits.

This validation framework was integrated into a wider **grid search** to systematically tune hyperparameters. We explored various recurrent architectures (RNN, GRU, LSTM), sequence lengths, and regularization techniques (e.g., L1/L2, Dropout, gradient clipping). Finally, for submitting predictions, we aggregate the classifications from all sequences of a test subject and use a **majority vote (mode)** to determine a single label for each subject.

4 Experiments

Baseline and Architecture Exploration

We first established baselines with a wide range of recurrent architectures, including standard **RNNs**, **LSTMs**, and **GRUs**. We tested both unidirectional and bidirectional configurations. These models were quite effective, with bidirectional GRUs in particular achieving strong validation F1-scores around **0.93-0.94**. However, they seemed to hit a performance ceiling at this level. Despite extensive tuning, we found they still struggled to perfectly distinguish the more subtle *low_pain* and *high_pain* classes, indicating we needed a model with either more representational power or a better regularization strategy to advance to the next level.

Table 2: Performance comparison of key models on our validation set.

Model	Accuracy	Precision	Recall	F1 (weighted)
Baseline (Bi-GRU)	93.45%	93.51%	93.45%	0.9347
Bi-LSTM (Weighted Loss)	94.22%	94.25%	94.22%	0.9423
Final Model (Bi-LSTM)	**98.12%**	**98.13%**	**98.12%**	**0.9811**

Data and Loss Function Experiments

We then explored two primary avenues for improvement. First, we experimented with **data enrichment**. We added measures of velocity and acceleration of the joint data and also analyzed the time-series **autocorrelation** as potential features. Neither approach provided a consistent signal that improved the F1-score, so they were discarded.

Second, we directly addressed the **class imbalance** by implementing the weighted Cross-Entropy Loss (equation 3) and also tested **Label Smoothing** as a regularization technique, but we ultimately found that a standard Cross-Entropy Loss combined with strong regularization produced a more stable and better performing model.

Final Model and Ensemble

Our most successful experiments involved a **bidirectional LSTM** architecture.

Through a grid search, we optimized the model's hyperparameters, reaching our best F1-score yet.

This model, with its **large number of parameters** ($\approx$ 8.5 million), proved highly effective once combined with the right techniques to avoid overfitting.

Finally, we experimented with an **ensemble** strategy. This involved training multiple models and averaging their output probabilities at inference time. However, this approach did not ultimately outperform our single Bi-LSTM model.

Figure 2: Training and validation loss (left) and F1-score (right) for our final Bi-LSTM model.

5 Results

As shown in Table 2, our final model, a 2 layer bidirectional LSTM with $\approx$ 8.5 million parameters, achieved a **weighted F1-score of 0.9811** on our held out, subject wise validation set.

This result was achieved by finding an effective **tradeoff between high model capacity and strong regularization**. Our data preprocessing and sequencing pipeline was essential, but we found that more advanced strategies like data enrichment did not provide a significant boost.

Interestingly, handling the class imbalance with a weighted loss function yielded a slightly lower F1-score (0.976) than a standard loss with a robust combination of L1, L2, and Dropout regularization.

6 Discussion

Our model's primary strength is its high generalization performance on the validation split, which we attribute to using a high capacity architecture.

The main challenge was the severe class imbalance. Counter intuitively, our experiments showed that strong regularization was a more effective strategy than explicit class-weighting in the loss function, which tended to destabilize training.

A limitation is this reliance on high capacity, which creates a significant overfitting risk that must be carefully managed. Additionally, our majority vote aggregation assumes a single pain state per subject, an assumption that might not capture entirely the pain dynamics.

7 Conclusions

We successfully tackled the classification pain states using a regularized Bi-LSTM and simple techniques for data preprocessing. However we could implement further improvements such as incorporating an **attention mechanism** to help the model focus on the most critical time steps.